

Isabella V. Garcia

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EDUCATION

University of California, San Diego

B.S., Bioengineering: Bioinformatics; Minor: General Biology

07/2022 – 06/2026

GPA: 3.71

SKILLS

Python/R/Java/C++ Proficiency

Git / GitHub

Unix systems, bash, remote development, HPC

Algorithms & Data Structures

Statistics & Probability

Machine learning

Data visualization & analysis

Pipeline development & usage

NGS sequence analysis

TECHNICAL EXPERIENCE

UC San Diego Bioengineering Department Senior Design Project

07/2025 – 06/2026

- Completed Bioengineering Department's Senior Design program under Dr. Max Qian at the J. Craig Venter Institute (JCVI)
- Developed software as part of a team for machine learning analysis and processing of flow cytometry data to classify leukemia and predict patient outcomes
- Conducted extensive literature review and presented findings weekly via formal reports
- Conducted analysis of ethics, safety, design considerations, and engineering standards

Genentech – Outstanding Student Awardee Intern

06/2025 – 09/2025

- 3-month Outstanding Student Awardee internship in the Media & Process Group, Cell Culture and Bioprocess Operations
- Researched and led development and implementation of a software tool to perform novel metabolic analyses, leveraging historical data and literature-based approaches to improve bioprocess operations
- Presented work to the group and held a poster session at the Intern Project Showcase

Cabrales Lab @ UC San Diego – Undergraduate Research Assistant

10/2023 – 06/2025

- Research assistant in the Functional Cardiovascular Engineering Research Group under PI Pedro Cabrales
- Led software development on hyperspectral characterization of hemoglobin-based oxygen carriers (HBOCs) using a microfluidic chip
- Led rapid low-cost %HbS quantification assay project
- Presented research poster at Spring 2024 Undergraduate Research Symposium

COMPLETED RELEVANT COURSEWORK

- BENG 181 - Molecular Sequence Analysis
- BENG 182 - Biological Databases
- BENG 183 - Applied Genomic Technologies
- CSE 151A - Machine Learning: Learning Algorithms
- CSE 185 - Advanced Bioinformatics Laboratory